

Distributions — Center, Spread, and Shape

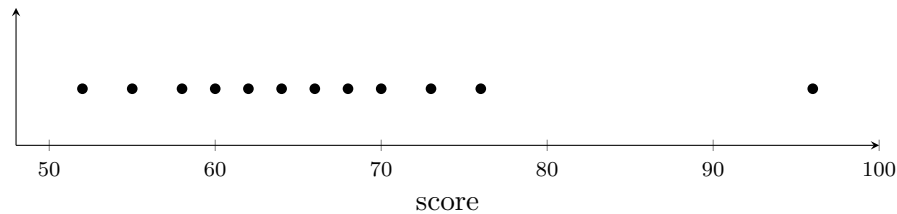
Choosing and computing the right summary of center and spread once the shape of the distribution is known

Directions:

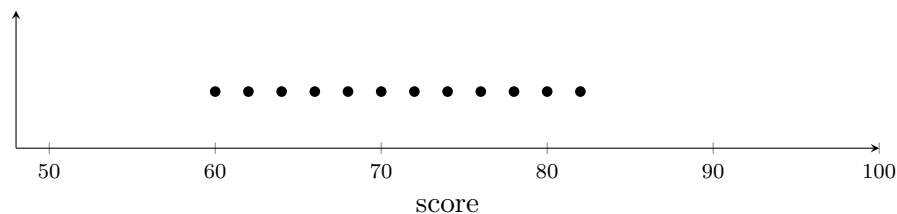
- Three displays are printed below: two dot plots of quiz scores and one histogram of commute times. Every problem names the display it uses.
- Use the school method for quartiles: Q_1 is the median of the lower half of the ordered data and Q_3 is the median of the upper half, with the overall median excluded when the count is odd.
- Round decimal answers to the nearest hundredth unless a problem says otherwise.
- Where a problem asks for a description of **shape**, the words are *symmetric*, *skewed right*, *skewed left*, and *outlier*.

The three displays

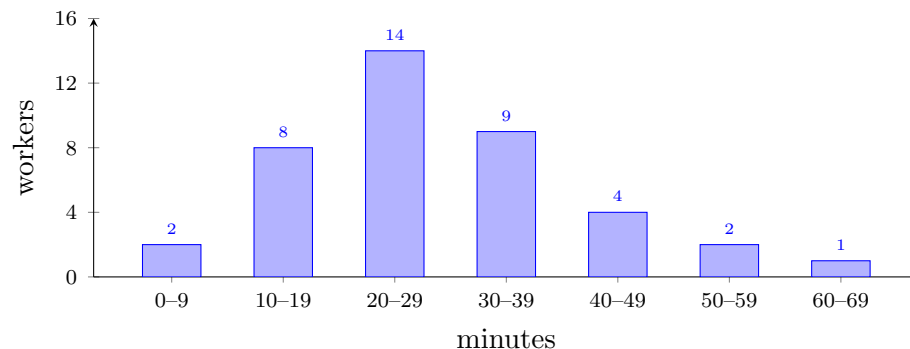
Class A quiz scores ($n = 12$): 52, 55, 58, 60, 62, 64, 66, 68, 70, 73, 76, 96



Class B quiz scores ($n = 12$): 60, 62, 64, 66, 68, 70, 72, 74, 76, 78, 80, 82



Commute times, in minutes, for a sample of workers:



1. Class A — the median. Find the median quiz score for Class A.

The data is already in order, so say which two values you averaged and why those two.

Answer: _____

2. Class A — the range. Find the range of the Class A scores.

Then name the single value that is doing most of the work in that answer.

Answer: _____

3. Class A — the mean. Find the mean quiz score for Class A, rounded to the nearest hundredth.

Show the total you divided by 12.

Answer: _____

4. Class A — quartiles and the IQR. Using the school method:

- (a) Find Q_1 , the median of the six lowest scores.
- (b) Find Q_3 , the median of the six highest scores.
- (c) Find the interquartile range, $IQR = Q_3 - Q_1$.

$Q_1 =$ _____ $Q_3 =$ _____ $IQR =$ _____

5. Class A — what the two centres tell you about the shape. You now have the mean (66.67) and the median (65) for Class A.

- (a) Write $>$, $<$, or $=$ between them: mean _____ median.
- (b) A long tail pulls the mean toward it and leaves the median where it is. Use your comparison and the dot plot to name the shape of the Class A distribution.

Answer: _____

6. Class B — a symmetric distribution.

- (a) Find the mean of the Class B scores. (b) Find the median.

mean = _____ median = _____

- (c) Compare the two answers, then say what that agreement tells you about the shape of Class B — and why it happened here but not for Class A.

7. Two measures of spread, disagreeing.

- (a) Find the range of the Class B scores. (b) Find the IQR of the Class B scores.

Class B range = _____ Class B IQR = _____

- (c) Class A's range was 44 and its IQR was 12.5; Class B's answers are above. Put the two ranges side by side, then the two IQRs. One pair says the classes are very different and the other says they are nearly the same. Explain which measure is telling the truth about the bulk of the data, and why the other one disagrees.

8. The commute histogram — reading the shape.

(a) Which interval holds the most workers? (b) How many workers are in the sample altogether?

modal interval _____ total workers _____

(c) Describe the shape of this distribution in one sentence, naming which side the tail is on and what that predicts about the mean compared with the median.

9. The commute histogram — a tail probability. One worker is chosen at random from the sample.

Find the probability that the worker's commute is at least 40 minutes. Give the answer as an exact fraction in lowest terms.

Show which bars you added.

Answer: _____

10. Challenge — reporting the right summary. A school newsletter has room for exactly one measure of centre and one measure of spread for each of the three distributions on this sheet.

For **Class A**, **Class B**, and the **commute times**, choose either “mean and standard deviation” or “median and IQR”, and justify each choice from the shape of that distribution. Your justification must say what the shape is and what would go wrong with the other choice — not simply which one you picked.

Answer: _____